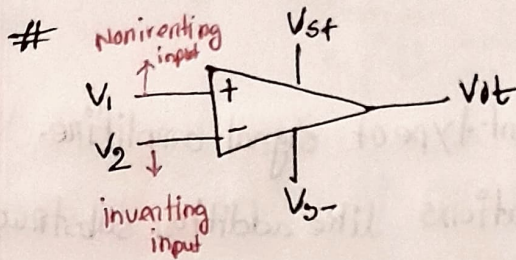


OP AMS

Operational amplifier

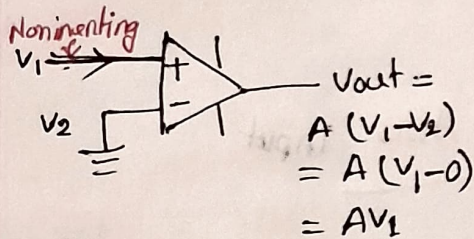
weak signal $\rightarrow$ Amplifier $\rightarrow$ strong signal

High gain amplifier



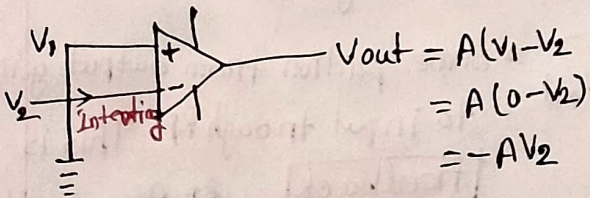
$$V_{out} = (V_1 - V_2) \text{ (A)} \rightarrow \text{Amplification}$$

Inverting/Non-inverting and their gain?



$$\begin{aligned} V_{out} &= A(V_1 - V_2) \\ &= A(V_1 - 0) \\ &= AV_1 \end{aligned}$$

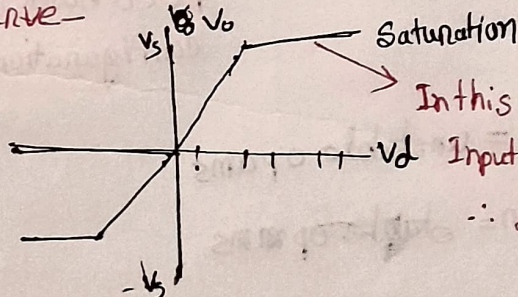
Input signal $\rightarrow$ [Sine wave]
Output signal $\rightarrow$ [Sine wave]
Input and output have same phase. No change. Only Amplification.



$$\begin{aligned} V_{out} &= A(V_1 - V_2) \\ &= A(0 - V_1) \\ &= -AV_1 \end{aligned}$$

Input signal $\rightarrow$ [Sine wave]
Output signal $\rightarrow$ [Inverted sine wave]
Input and output have different phase. Output will be (-) from of input and amplify.

Voltage transfer curve



In this point supply voltage =
Input voltage so saturated.
 $\therefore$ supply voltage $>$ input voltage

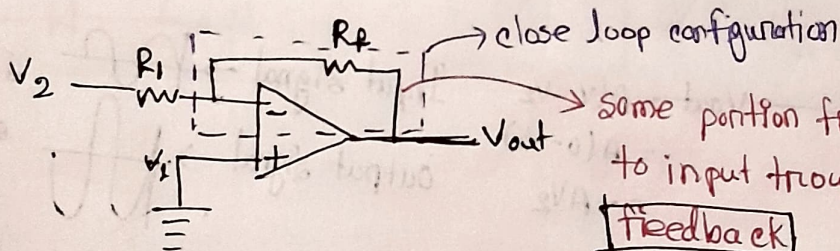
characteristic -

why to use -

- i) can be configured to work as different type of signal amplifier.
- ii) Use to perform mathematical operations like addition subtraction.
- iii) can be use to constructed active filters and oscillator.
- iv) can be use as comparator.

Inverting Op Ams input

mostly use for constant gain

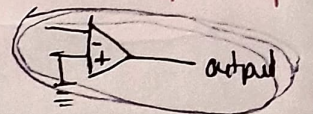


close loop configuration
some portion from output going to input through R_f . This is

feedback

R_f = feedback resistor

feedback configuration, ~~not~~ open loop



~~# open loop~~

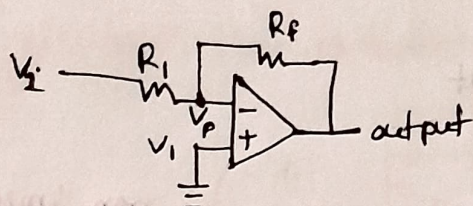
open loop $\Rightarrow$ high gain = unstable opams

close loop $\Rightarrow$ less gain = stable opams

* Open loop opams এ gain এর পরিমাণ হালকা তবুও unstable situation তৈরি হতে পারে. close loop এ তাহলে feedback resistor resistor ফিডব্যাক voltage loose করে And stable situation এ থাকে.

R_1 resistor use $\rightarrow$ to distinguish between input and feedback input.

Virtual ground



$V_p \ll V_{out}$ so can consider it zero.

this is called virtual ground (V_p)

$$i(\text{input}) = \frac{V_i - V_p}{R_1} = \frac{V_{out} - V_p}{R_f}$$

$$\Rightarrow \frac{V_i}{R_1} = \frac{-V_{out}}{R_f}$$

$$\frac{V_i}{R_1} = -\frac{V_{out}}{R_f}$$

$$\Rightarrow \frac{V_{out}}{V_i} = -\frac{R_f}{R_1}$$

$$\Rightarrow A = -\frac{R_f}{R_1}$$

$$\Rightarrow A = -\frac{R_f}{R_1} = -\frac{V_{out}}{V_i}$$

$\rightarrow$ this ~~not~~ inverts the input 180°

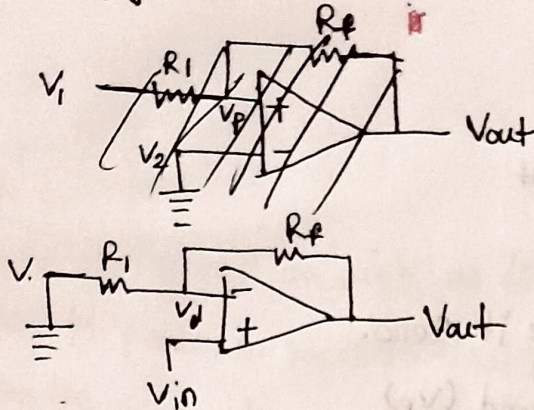
We know

$$V_{out} = A V_i$$

$$\therefore A = \frac{V_{out}}{V_i}$$

NON-Inverting opAMS input

Constant gain # can use as voltage follower or unit gain buffer.

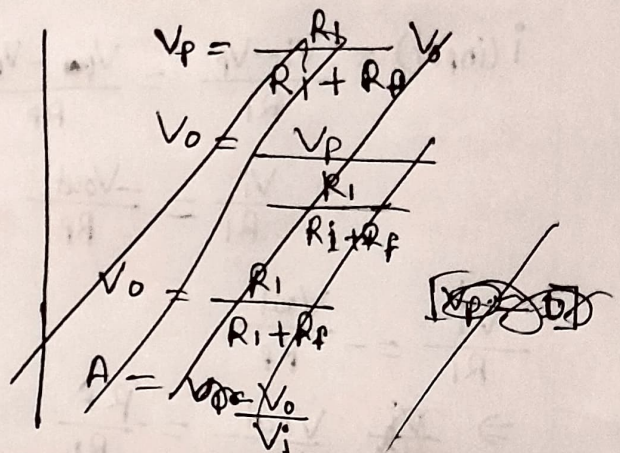
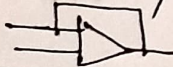


Same as ~~non~~ inverting but
Amplification + Same face

$$\frac{V_o}{V_i} = 1 + \frac{R_f}{R_i}$$

$$A = \frac{V_o}{V_i} = 1 + \frac{R_f}{R_i}$$

$R_f = 0$
 $R_i = \infty$ $\rightarrow A = 1 \rightarrow$ unity gain buffer



$$V_p = \frac{R_i}{R_i + R_f} V_o$$

$$V_o = -A(V_i - V_p)$$

$$= A\left(V_i - \frac{R_i}{R_i + R_f} V_o\right)$$

$$V_o = A V_i - \frac{A R_i}{R_i + R_f} V_o$$

$$A V_i = V_o + A \frac{R_i V_o}{R_i + R_f}$$

$$A V_i = A V_o \left(1 + \frac{R_i}{R_i + R_f}\right)$$

$$\Rightarrow V_o = \frac{A V_i}{1 + \frac{R_i}{R_i + R_f}}$$

$$V_o = \frac{A V_i}{1 + \frac{A R_i}{R_i + R_f}}$$

$A \gg \gg 1$
1 neglect

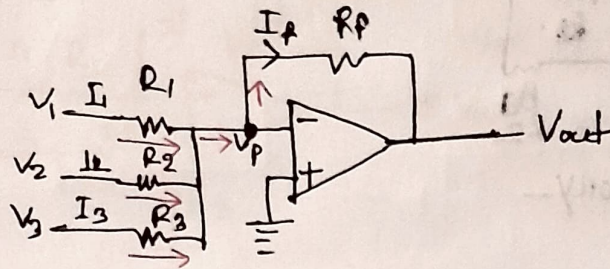
$$\frac{V_o}{V_i} = \frac{R_i}{R_i + R_f}$$

$$= 1 + \frac{R_f}{R_i}$$

$$A = 1 + \frac{R_f}{R_i}$$

Inverting Adder op-ams

An op-ams circuit that can combine number of input signal to a signal output that is weighted sum of applied voltage



$$V_p = 0V$$

$$i_1 = \frac{V_1 - V_p}{R_1}; \quad i_2 = \frac{V_2 - V_p}{R_2}; \quad i_3 = \frac{V_3 - V_p}{R_3}; \quad i_f = \frac{V_p - V_{out}}{R_f}$$

$$K.C.L \Rightarrow$$

$$i_1 + i_2 + i_3 = i_f$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_{out}}{R_f} \quad [V_p = 0]$$

$$\Rightarrow V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

$$= - \left(\frac{V_1 R_f}{R_1} + \frac{V_2 R_f}{R_2} + \frac{V_3 R_f}{R_3} \right) \quad [\text{Scaling}]$$

When $R_f \neq R_1 \neq R_2 \neq R_3$ — case 1

$$V_o = -(V_1 + V_2 + V_3)$$

When $R_f = R_1 = R_2 = R_3$ [Summing]

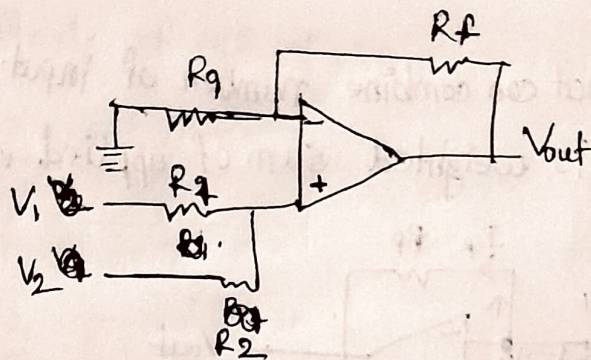
— case 2

$$V_o = -\frac{R_f}{R_1} (V_1 + V_2 + V_3) \quad [\text{Avg}] \quad \text{— case - 3}$$

$$= -\frac{1}{n} (V_1 + V_2 + V_3) \quad \text{When } R_1 = R_2 = R_3 \text{ and } \frac{R_f}{R} = \frac{1}{n}$$

↘ number of

Non-Inverting Adder



$$V_{out} = \left(1 + \frac{R_f}{R_g}\right) V^+$$

Superposition theory -

If, $V_2 = 0$,

$$V_1 = \left(\frac{R_2}{R_1 + R_2}\right) V_1$$

V_0

If $V_1 = 0$,

$$V_2 = \left(\frac{R_1}{R_1 + R_2}\right) V_2$$

$$V_{out}^+ = V_1 + V_2$$

$$= \frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2$$

$$\therefore V_{out} = \left(1 + \frac{R_f}{R_g}\right) \left(\frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2\right)$$

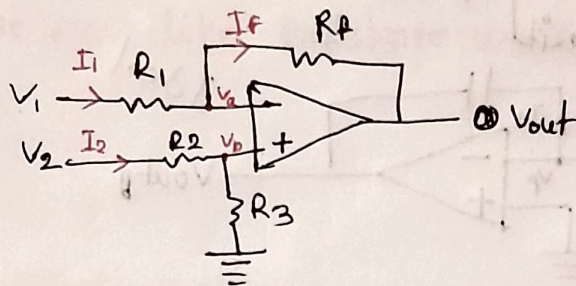
$$V_{out} = \left(1 + \frac{R_f}{R_g}\right) (V_1 + V_2) \quad [\text{when } R_1 = R_2 = R_g]$$

Subtractor

simple opams also a differential amplifier.

But voltage gain is high. so we use subtractor

→ close loop



[combination of inverting and non inverting amplifier]

~~Inverting~~

Inverting $\Rightarrow V_o = -\left(\frac{R_f}{R_i}\right) V_i$

Non-inverting $\Rightarrow V_o = \left(1 + \frac{R_f}{R_i}\right) V_{in}$

$$I_1 = \frac{V_1 - V_a}{R_1} ; I_2 = \frac{V_2 - V_b}{R_2} ; I_f = \frac{V_b - V_{out}}{R_f}$$

If $V_2 = 0$

$$V_{out(a)} = -V_1 \left(\frac{R_f}{R_1} \right)$$

If $V_1 = 0$

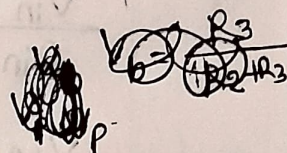
$$V_{out(b)} = V_2 V_b \left(\frac{R_f}{R_1} \right)$$

$$= V_2 \left(\frac{R_f}{R_1} \right) \left(\frac{R_3}{R_2 + R_3} \right)$$

$\therefore V_{out} = V_{out(a)} + V_{out(b)}$

$$= -V_1 \left(\frac{R_f}{R_1} \right) + V_2 \left(\frac{R_3}{R_2 + R_3} \right) \left(\frac{R_1 + R_f}{R_1} \right)$$

Use as → subtractor
comparator
high activation switch



$$V_b = \left(\frac{R_3}{R_2 + R_3} \right) V_2$$

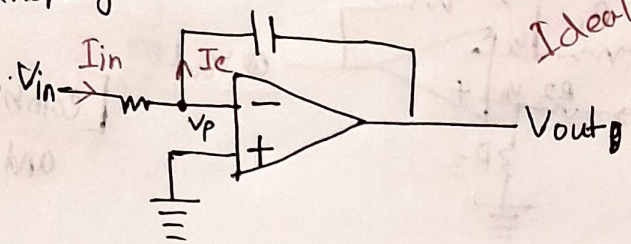
Superposition theory

Integrator

instade of feed back resistor we use capacitor.

Analog computer
A/D converter
wave shaping circuit

→ use



K.C.L ⇒

$$I_{in} = I_e$$

$$\frac{V_{in} - V_p}{R_{in}} = I_e$$

$$Q = C V_c$$

$$\frac{dQ}{dt} = C \frac{dV_c}{dt}$$

$$I_e = C \frac{dV_c}{dt}$$

$$\frac{V_{in}}{R_{in}} = C \frac{dV_c}{dt}$$

$$\frac{V_{in}}{R_{in}} = C \frac{d(V_p - V_{out})}{dt} \quad [V_p = 0]$$

$$\frac{V_{in}}{R_{in}} = -C \frac{dV_{out}}{dt}$$

$$\Rightarrow -\frac{V_{in}}{R_{in} C} dt = dV_{out}$$

$$\Rightarrow V_{out} = \left[-\frac{1}{R_{in} C} \right] \int V_{in} \quad [\text{integrated both side}]$$

→ scaling factor

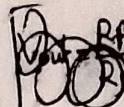
TKS  website

Capacitor Reactance $(X_c) = \frac{1}{\omega c}$

As like Resistance but change with time

$$= \frac{1}{2\pi f c}$$

$$V_{out} = -\left(\frac{R_f}{R_i}\right) V_{in}$$

As X_c are most like Resistance we can say, $V_{out} = -\frac{X_c}{R_{in}} V_{in}$ 

$\therefore$ gain $A = -\frac{X_c}{R_{in}}$

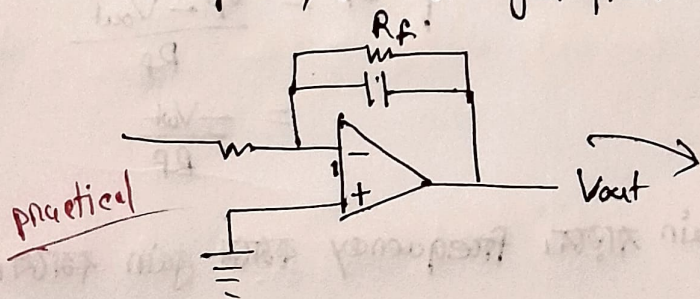
$$= \frac{1}{2\pi f c R_{in}}$$

Open Loop



limitations $\rightarrow$

- ① high gain / unstable situation
- ② at frequency $\rightarrow$ gain provide error voltage (offset voltage)

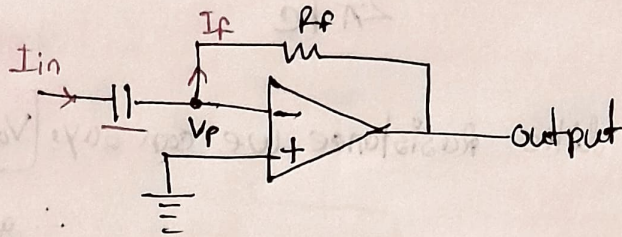


To solve the limitation of A due to X_c

$$\therefore A = -\frac{R_f}{R_{in}}$$

Differentiator

Instead of R_2 resistor we use capacitor.



Ideal

K.C.L $\Rightarrow I_{in} = I_f$

$$\Rightarrow \frac{-V_{out}}{R_f} = C \frac{dV_{in}}{dt}$$

$$\Rightarrow \boxed{V_{out} = -C R_f \frac{dV_{in}}{dt}}$$

$$I_{in} = C \frac{dV_{in}}{dt}$$

$$= C \frac{d(V_{in} - V_p)}{dt}$$

$$= C \frac{dV_{in}}{dt}$$

$$I_f = \frac{V_p - V_{out}}{R_f}$$

$$= \frac{-V_{out}}{R_f}$$

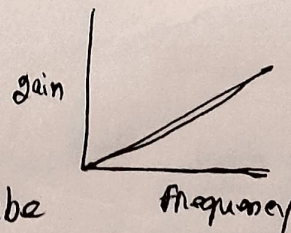
frequency বাড়লে gain বাড়বে, frequency কমলে gain কমবে।

$$\Rightarrow A = -\frac{R_f}{X_c} V_{in} \quad \left[-\frac{R_f}{X_c} V_{in} \right]$$

$$X_c = \frac{1}{\omega C}$$

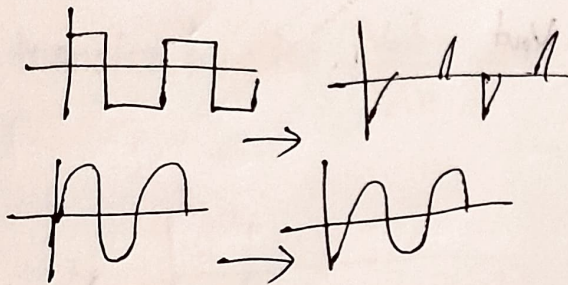
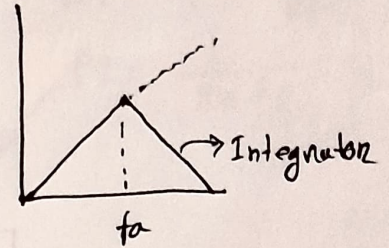
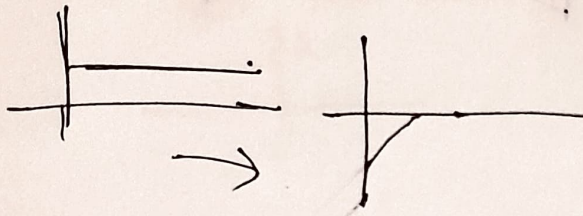
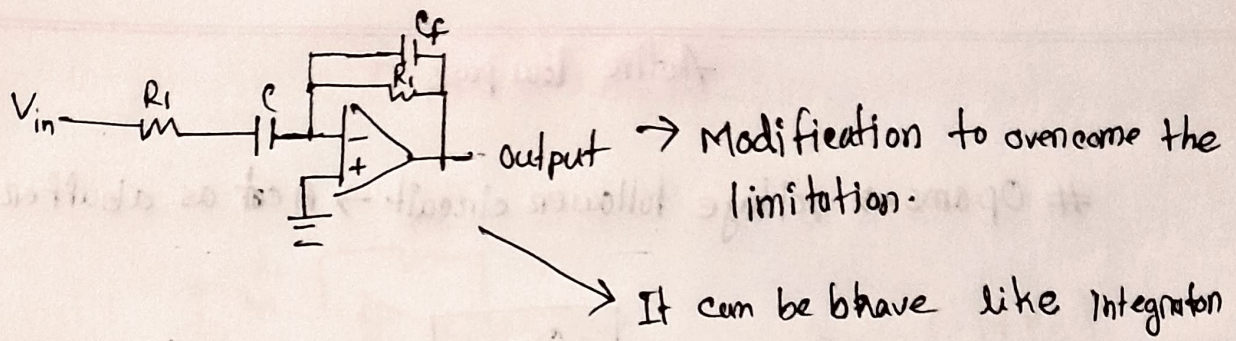
$$= -R_c \omega C V_{in}$$

Limitation



- ① higher frequency to circuit unstable
হয় এবং oscillate করে

- ② capacitor use করার জন্য noise dominant input signal আসবে



$$V_{out} = \frac{1}{R_1 C_f} \int V_{in} dt$$

$$V_{out} = \frac{V_{in}}{R_1 C_f} \int dt$$

$$V_{out} = \frac{V_{in}}{R_1 C_f} t$$

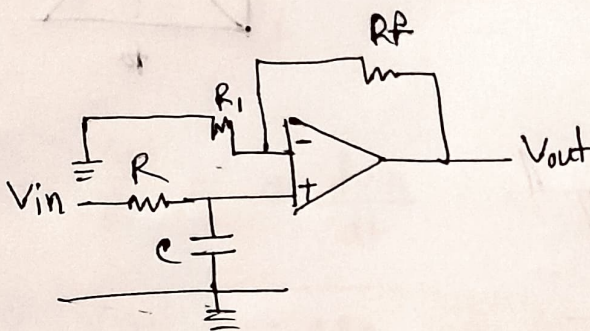
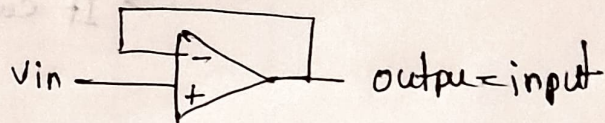
$$V_{out} = \frac{V_{in}}{R_1 C_f} \int dt$$

$$V_{out} = \frac{V_{in}}{R_1 C_f} t$$



Active low pass

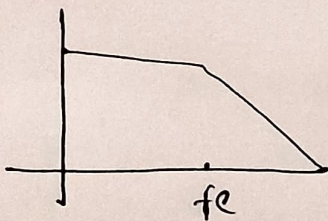
Op amps $\rightarrow$ voltage follower circuit $\rightarrow$ Act as a buffer



$$A = \left(1 + \frac{R_F}{R_1}\right)$$

cut off voltage $\Rightarrow f_c = \frac{1}{2\pi RC}$ Hz

$$\frac{V_{out}}{V_{in}} = \frac{A_v \times 1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

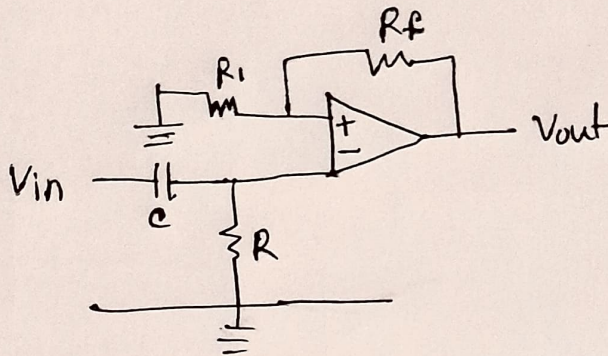


$$V_{out} = \frac{X_c}{X_c + R} \times V_{in} \quad \text{[for low pass \& passive filter]}$$

$$\frac{V_{out}}{V_{in}} = \frac{X_c}{\sqrt{R^2 + X_c^2}} = \frac{\frac{1}{\omega C}}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$$

$$= \frac{1}{\sqrt{R^2 \omega^2 C^2 + 1}} \quad \frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}} = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

Active High pass filter



$$A_v = \left(1 + \frac{R_f}{R_i} \right)$$

$$f_c = \frac{1}{2\pi RC}$$

cut-off $\swarrow$

transfer function $\frac{V_{out}}{V_{in}} = A_v \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}}$

